

WHEN DOES SYMMETRIC SOURCE PAIRING HELP FLOW POLICIES FOR ROBOT CONTROL?

Anonymous authors

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ABSTRACT

Flow-matching robot policies generate actions by integrating a learned vector field from a source sample, making test-time source selection part of the deployed controller. We study a simple, training-free rule that averages the two action endpoints produced from a Gaussian source and its exact negation. At equal total neural-function evaluations, this symmetric projection recovers 0.945 ± 0.059 return (mean $\pm$ SD across four independently trained Unitree Go2 policies) relative to the official random-source rule; the policy-level 95% Student- t interval is $[0.852, 1.038]$, while the conservative exact one-sided sign-test value is 0.0625. Across the same policies, symmetric pairing also exceeds two-source IID averaging by 0.566 return at equal pair compute. Mechanism measurements show near-opposite source-conditioned action displacements (mean cosine -0.969) and a residual RMS ratio of 0.124, versus 0.710 for IID averaging. A batched implementation is $1.19\times$ faster than sequential pairing and, at batch 4096 on an H200, is faster than one 64-step random solve. Strong controls delimit the result: deterministic zero-source inference matches or exceeds symmetric pairing, the symmetry-specific advantage does not resolve on Spot, and mirrored closed-loop rollouts fail a preregistered variance-reduction gate. Symmetric pairing is therefore a reliable correction for stochastic-source degradation on these Go2 policies, not a universally superior deployment rule.

1 INTRODUCTION

Flow and diffusion policies represent complex action distributions through an iterative generative process rather than a simple parametric likelihood (Chi et al., 2023; Lipman et al., 2022). Flow Policy Optimization (FPO) and FPO++ extend this representation to online policy-gradient training without exact action likelihoods (McAllister et al., 2025; Yi et al., 2026). Their flexibility introduces a deployment choice that does not arise for a deterministic policy: from which source should the action flow be integrated? The official FPO++ evaluation uses either a fresh Gaussian source or a zero source, and reports that the latter often performs better. This gap matters scientifically because it reveals that part of the learned stochastic action variation degrades return, and practically because source selection changes both behavior and solver cost without retraining.

We ask a narrow question: can a symmetric source construction remove the harmful component of stochastic inference while preserving a controlled amount of flow-policy averaging? Given the endpoint map $F_\theta(o, z)$ for observation o and source z , we evaluate

$$a_{\text{sym}}(o, z) = \frac{1}{2} [F_\theta(o, z) + F_\theta(o, -z)]. \quad (1)$$

This is the classical antithetic construction applied to conditional policy endpoints. Antithetic sources and approximate affine antisymmetry are already established for generative models (Jia et al., 2025); our goal is not to reintroduce them. Instead, we determine what survives in closed-loop robot control under equal-compute, deterministic, IID-averaging, cross-policy, and cross-task controls.

The central result is deliberately qualified. Across four independently trained Go2 policies, symmetric projection consistently recovers the official random-source return deficit at equal total solver evaluations. Direct endpoint measurements connect this recovery to cancellation of a dominant source-odd action displacement. However, zero-source inference is at least as strong, and on Spot the data do not separate symmetric pairing from generic two-endpoint averaging. A further attempt to use mirrored

Table 1: Frozen-policy inference rules. $z, w \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I)$ and F^K denotes an Euler solve with K steps.

Rule	Action	Endpoints	Total NFE
zero64	$F^{64}(o, 0)$	1	64
random64	$F^{64}(o, z)$	1	64
IID-pair32	$[F^{32}(o, z) + F^{32}(o, w)]/2$	2	64
symmetric32	$[F^{32}(o, z) + F^{32}(o, -z)]/2$	2	64
IID-pair64	$[F^{64}(o, z) + F^{64}(o, w)]/2$	2	128
symmetric64	$[F^{64}(o, z) + F^{64}(o, -z)]/2$	2	128

sources as a closed-loop return control variate fails. The evidence supports a robust diagnostic and correction for one stochastic-inference failure mode, not a new state of the art.

Our contribution is a controlled account of this phenomenon: (i) policy-level replication across four Go2 training seeds with equal-total-NFE and equal-pair-NFE comparisons; (ii) a source-parity mechanism audit and a batched throughput implementation; and (iii) deterministic, Spot, and mirrored-rollout controls that expose where the appealing local symmetry does not translate into a stronger deployment policy. All analyses read archived evaluation records, and the appendix maps every reported number to a committed artifact.

2 BACKGROUND AND INFERENCE RULES

2.1 FLOW POLICIES

Flow matching learns a time-dependent vector field that transports a simple source distribution to a target data distribution (Lipman et al., 2022; Liu et al., 2022). In a conditional robot policy, the vector field is conditioned on observation o . Starting at $x_0 = z \sim \mathcal{N}(0, I)$, an ODE solver produces the endpoint $F_\theta(o, z) = x_1$, which is interpreted as the action. FPO replaces explicit action-likelihood ratios with ratios derived from conditional flow-matching losses, while FPO++ adds per-sample ratios and an asymmetric trust region to stabilize robot-control training (McAllister et al., 2025; Yi et al., 2026). These training details remain unchanged in our study: every comparison uses a frozen official FPO++ policy.

2.2 CONTROLLED SOURCE SELECTION

Table 1 defines the four inference rules. Random and zero use one endpoint. IID-pair and symmetric inference average two endpoints, differing only in whether the second source is independent or the exact negative of the first. We count one vector-field evaluation per Euler step and explicitly match total NFE when making compute claims. In particular, two 32-step symmetric endpoints and one 64-step random endpoint both cost 64 NFE per action. The equal-pair attribution comparison uses 64 steps for both endpoints in both IID and symmetric methods, for 128 total NFE.

2.3 SOURCE-PARITY DIAGNOSTIC

Let $D(o, z) = F_\theta(o, z) - F_\theta(o, 0)$ be the source-conditioned displacement from the zero-source endpoint. Its even and odd components are

$$D_{\text{even}}(o, z) = \frac{1}{2}[D(o, z) + D(o, -z)], \quad D_{\text{odd}}(o, z) = \frac{1}{2}[D(o, z) - D(o, -z)]. \quad (2)$$

Symmetric projection leaves only D_{even} . We report the cosine between $D(o, z)$ and $D(o, -z)$ and the normalized residual $\|D_{\text{even}}\|_{\text{RMS}}/\|D(o, z)\|_{\text{RMS}}$. A small residual indicates an approximately affine, source-antisymmetric endpoint map. The corresponding IID ratio replaces $-z$ with an independent w and tests whether cancellation is specific to symmetry rather than averaging alone.

3 EXPERIMENTAL DESIGN

3.1 TASKS, POLICIES, AND PAIRED EVALUATION

We use the official IsaacLab FPO++ locomotion setup (Yi et al., 2026). Go2 experiments train four policies with seeds 42–45, each for 1500 iterations with 4096 parallel environments and the released hyperparameters. We evaluate the fixed final checkpoint `model_1499.pt`; no run is excluded and no checkpoint is selected by evaluation return. The primary independent unit for cross-policy claims is the training seed, not the thousands of nested simulator episodes.

Before changing inference, the official seed-42 run reproduced the reported approximately 40-return regime: final training return was 40.620, while the fixed final checkpoint obtained 41.529 with zero-source evaluation and 40.523 with random-source evaluation. The runner completed all 1500 training iterations and evaluated all 31 saved checkpoints with 4096 episodes per source mode and checkpoint. Subsequent method comparisons use fixed final checkpoints rather than selecting the best evaluation iteration.

Within a policy, comparisons are paired by initial environment state. Random, IID-pair, and symmetric modes share the complete primary source stream; IID uses a separately seeded secondary stream. The three new Go2 policies use 512 episodes per method for the 128-NFE attribution evaluation and 256 episodes per method for the equal-total-NFE evaluation. Seed 42 uses the corresponding locked 256-episode studies. All validity gates require equal initial-observation hashes, expected endpoint counts and NFE, finite returns, and unchanged actor parameters.

For cross-task evaluation, we train one official Spot seed-42 policy for 1500 iterations and evaluate its fixed final checkpoint in five paired cells: zero64, zero32, random64, IID-pair32, and symmetric32. Each cell contains 256 episodes, for 1280 valid episodes total. This experiment was designed as an attribution test: transfer requires both a positive symmetric-minus-random interval and a positive symmetric-minus-IID interval.

3.2 STATISTICS

For generalization across Go2 policies, we compute the paired return difference within each training seed and summarize the four seed effects using their mean, sample SD, and a two-sided 95% Student- t interval with three degrees of freedom. Because four seeds are too few to assess normality, we additionally report the exact one-sided sign-test value. Four same-direction effects yield $p = 1/16 = 0.0625$; we do not describe this as conventionally significant. Within-policy comparisons use the preregistered paired bootstrap with 20,000 resamples. Spot intervals likewise resample paired episode indices. The mirrored-rollout variance ratio uses 20,000 paired bootstrap resamples. All interval definitions and statistical units are stated with their results.

4 RESULTS

4.1 SYMMETRIC PROJECTION RECOVERS THE GO2 RANDOM-SOURCE DEFICIT

Figure 1A shows the equal-total-NFE comparison. Symmetric32 minus random64 is positive for all four policies: 0.981, 0.903, 1.008, and 0.888. The mean effect is 0.945 return with SD 0.059 and policy-level 95% t interval $[0.852, 1.038]$. The magnitude is stable across independently trained policies, although the exact sign-test value remains 0.0625 because $n = 4$.

The stronger attribution comparison holds pair compute fixed. Symmetric64 minus IID-pair64 is also positive for all four policies, with effects 0.779, 0.357, 0.574, and 0.551. Its mean is 0.566 (SD 0.173), with policy-level interval $[0.291, 0.840]$ and exact one-sided sign-test $p = 0.0625$. Generic two-endpoint averaging therefore does not fully explain the Go2 recovery.

Deterministic deployment changes the conclusion. Symmetric32 minus zero32 has mean -0.023 (SD 0.044) and interval $[-0.094, 0.047]$. Thus symmetric projection repairs the official stochastic inference rule, but it does not improve on the stronger deterministic zero-source control.

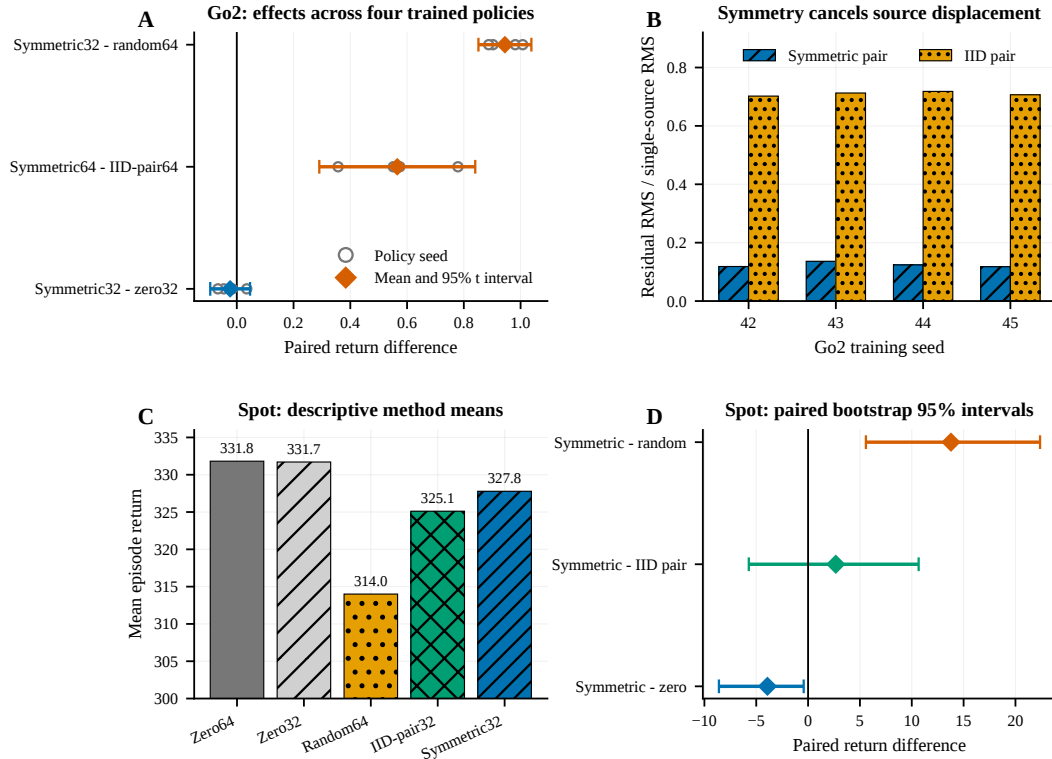


Figure 1: Core results and controls. **A**: paired Go2 return effects across four independently trained policies. Open circles are training-seed effects; diamonds and bars are the policy-level mean and two-sided 95% Student- t interval ($n = 4$ policies). The first and third comparisons use equal 64 total NFE; the second uses equal 128 total NFE. **B**: normalized endpoint residuals for exact symmetric and IID pairs on 128 environments over eight rollout steps per policy; lower is stronger cancellation. **C**: descriptive Spot means from five paired cells of 256 episodes. **D**: Spot paired return differences with 20,000-sample bootstrap 95% intervals. Symmetry recovers the random-source deficit on both tasks, but does not beat zero source and is not separated from IID averaging on Spot.

4.2 THE ENDPOINT MAP IS APPROXIMATELY AFFINE ANTISYMMETRIC

Across the same four policies, the cosine between $D(o, z)$ and $D(o, -z)$ is -0.969 on average (range -0.972 to -0.964). The symmetric residual ratio is 0.124 (range 0.118 – 0.136), whereas the IID residual ratio is 0.710 (range 0.702 – 0.718); Figure 1B shows that this gap is present for every policy. In absolute units, the even-component RMS is 0.0386 and the odd-component RMS is 0.3088 . The source effect is therefore dominated by an odd component that exact pairing removes. This evidence closely matches the affine-antisymmetry account reported for other generative models (Jia et al., 2025), now measured in conditional robot actions rather than images or uncertainty estimators.

4.3 SPOT SHOWS RECOVERY WITHOUT SYMMETRY-SPECIFIC ATTRIBUTION

On Spot, random64 obtains mean return 314.00 , IID-pair32 325.11 , symmetric32 327.78 , zero32 331.70 , and zero64 331.81 (Figure 1C). Symmetric32 exceeds random64 by 13.78 with paired bootstrap interval $[5.58, 22.37]$. The improvement relative to IID-pair32 is only 2.67 , with interval $[-5.72, 10.66]$, so the locked symmetry-specific transfer gate fails. Symmetric32 is also worse than zero32 by -3.92 , interval $[-8.60, -0.42]$. The cross-task result supports a broader two-endpoint averaging benefit relative to random inference, but cannot attribute that benefit specifically to exact source negation.

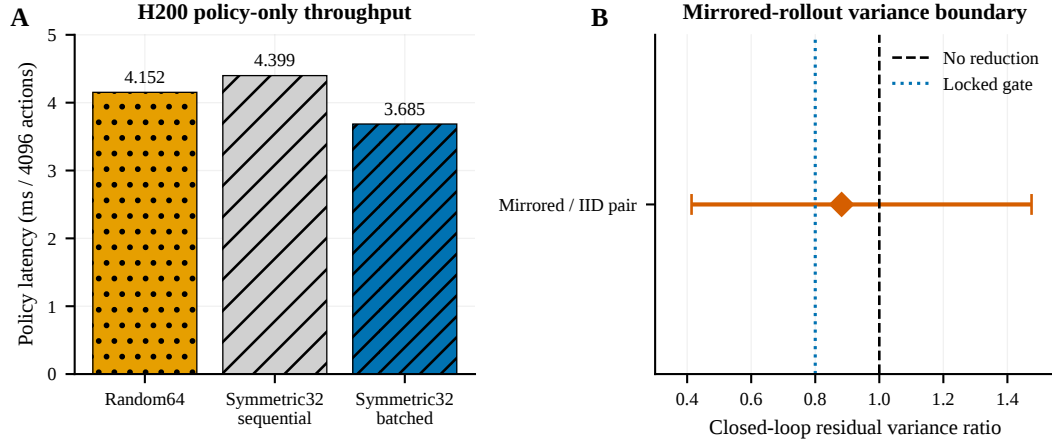


Figure 2: Efficiency and failure boundary. **A**: median policy-only latency for 4096 frozen Go2 observations on one H200, after 10 warm-ups and over five blocks of 20 synchronized calls. This is neither end-to-end simulator latency nor single-robot latency. **B**: mirrored-to-IID closed-loop pair-residual variance ratio with a 20,000-sample bootstrap 95% interval. The interval crosses the no-reduction line and the point estimate misses the locked 0.80 gate.

4.4 BATCHED INFERENCE REMOVES THE POLICY-ONLY LATENCY PENALTY

Concatenating observations and the $z/-z$ sources along the batch dimension computes both 32-step endpoints in one doubled policy call. On an H200 at batch 4096, random64 takes 4.152 ms, sequential symmetric32 takes 4.399 ms, and batched symmetric32 takes 3.685 ms (Figure 2A). Batching is $1.194\times$ faster than sequential pairing and uses $0.888\times$ the random64 latency in this high-parallelism policy-only measurement. Batched and sequential actions differ by at most 4.77×10^{-7} . The result shows that equal NFE can translate into favorable accelerator throughput; it does not establish latency on physical control hardware.

4.5 LOCAL SYMMETRY DOES NOT SURVIVE AS A RELIABLE RETURN CONTROL VARIATE

A natural extension is to execute z and $-z$ in separate environments and average their returns, as antithetic perturbations and common random numbers can reduce variance in optimization or rollout comparison (Salimans et al., 2017; Yu et al., 2018; Yadav et al., 2026). We preregistered a small Go2 test using 256 paired environments and compared mirrored and IID return residual variance. The mirrored/IID ratio is 0.882, with bootstrap interval $[0.414, 1.475]$ (Figure 2B), failing both the ratio-at-most 0.80 gate and the requirement that the upper bound be below one. The paired returns remain positively correlated (0.479 for mirrored versus 0.450 for IID). Immediate action antisymmetry is therefore not preserved by closed-loop dynamics strongly enough to support this training-control-variate extension.

5 RELATED WORK

Flow and diffusion policies. Flow matching trains continuous normalizing flows through vector-field regression (Lipman et al., 2022); rectified flow emphasizes straight transport paths and coarse integration (Liu et al., 2022). Generative policies apply related iterative samplers to multimodal robot actions (Chi et al., 2023). FPO introduces a conditional-flow-matching surrogate for policy optimization, and FPO++ makes it effective across locomotion, motion tracking, and manipulation (McAllister et al., 2025; Yi et al., 2026). We leave training untouched and isolate a test-time source-selection effect in frozen FPO++ policies.

Antithetic sampling and common randomness. Mirrored perturbations are widely used to reduce estimator variance, including evolution strategies and bandit gradients (Salimans et al., 2017; Yu et al., 2018). Common random numbers similarly improve comparisons between simulated rollouts under

appropriate coupling assumptions (Yadav et al., 2026). Most directly, Jia et al. (2025) show that $z/-z$ initial noises generate negatively correlated endpoints across diffusion and other generative models, and propose approximate affine antisymmetry as the mechanism. Our study adopts that established construction and asks a different question: whether endpoint averaging improves closed-loop robot return under equal-compute and deterministic controls. The Go2 result is positive, while the Spot attribution and mirrored-rollout experiments mark important limits.

6 LIMITATIONS

The principal limitation is breadth. The policy-level result contains four Go2 training seeds but only one Spot policy, and no H1, G1, manipulation, or physical-robot confirmation. Four same-direction Go2 effects are encouraging, yet the exact sign test cannot cross 0.05 with only four units. The Spot interval against IID averaging is wide and crosses zero, so cross-task evidence does not support a universal symmetry-specific effect.

The method is also not the best observed deployment rule. Zero-source inference matches symmetric projection on Go2 and exceeds it on Spot. The practical value is consequently diagnostic and conditional: symmetric projection preserves a stochastic construction while cancelling a harmful source-odd component, but applications that do not require stochastic source semantics should prefer the validated zero-source baseline. Throughput was measured on one H200 at batch 4096; doubled batching may not help at low batch size or on embedded hardware.

Finally, the mechanism evidence is empirical. Approximate affine antisymmetry explains immediate endpoint cancellation, but closed-loop feedback can alter the observation distribution, as demonstrated by the failed mirrored-return variance test. We do not claim a theorem connecting action-space parity to long-horizon return.

7 CONCLUSION

Symmetric source pairing reliably recovers the stochastic-inference deficit of flow-matching control policies across four independently trained Go2 agents at equal compute. Endpoint audits identify a stable source-odd displacement that exact pairing cancels, and accelerator batching makes the construction efficient at official training parallelism. Just as importantly, deterministic and cross-task controls prevent a broader conclusion: zero-source inference is stronger, Spot does not isolate a symmetry-specific advantage, and mirrored rollouts do not provide a reliable return control variate. These results turn a visually compelling generative-model symmetry into a bounded, reproducible robot-control finding rather than an overstated deployment method.

BROADER IMPACT

This work concerns simulated locomotion policy evaluation and does not introduce new physical-robot capability. More reliable source-selection diagnostics can reduce wasted evaluation compute and discourage selecting a stochastic inference rule without deterministic controls. Conversely, simulation returns and high-batch GPU timing should not be interpreted as physical safety or real-time deployment evidence. Any deployment should separately validate latency, stability, and failure recovery on its target hardware.

LLM USAGE DISCLOSURE

Language-model assistance was used to organize experiment records, draft text, and generate plotting and build scripts. All numerical claims are produced from committed experiment artifacts by deterministic scripts, and all cited metadata was checked against arXiv or Crossref records. The authors remain responsible for the claims, code, and final manuscript.

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A REPRODUCIBILITY DETAILS

A.1 ARTIFACT MAP

Table 2 maps each result family to its frozen protocol, analysis, and machine-readable record. All paths are relative to the repository root. The figure script reads these JSON records directly and does not contain hard-coded result values.

Table 2: Paper claims and committed artifact directories.

Claim	Artifact directory
Official Go2 reproduction	experiments/baseline-go2
Four-policy return effects	experiments/go2-antithetic-cross-policy-summary
Independent Go2 policies	experiments/go2-antithetic-independent-seeds
Equal-total-NFE seed 42	experiments/go2-antithetic-equal-total-nfe
Affine antisymmetry	experiments/go2-antithetic-affine-mechanism
H200 throughput	experiments/go2-antithetic-batched-throughput
Spot control	experiments/spot-antithetic-cross-task
Mirrored rollout boundary	experiments/go2-mirrored-rollout-variance

A.2 IMPLEMENTATION

The inference helper accepts explicit source tensors. IID-pair inference returns $[F(o, z) + F(o, w)]/2$; symmetric inference invokes the same helper with $(z, -z)$. The batched version concatenates (o, o) and $(z, -z)$, calls the unchanged policy once, splits the resulting endpoints, and averages them in float32. Validation checks source shape, floating-point type, finite values, endpoint shape, and the expected doubled batch dimension. Unit tests verify exact negation, arithmetic averaging, non-finite rejection, and single-call batching.

A.3 EVALUATION CONFIGURATION

All locomotion training uses 4096 environments and 1500 iterations. Evaluations use one completed episode per environment. Go2 independent-policy attribution uses 512 environments per method for seeds 43–45; equal-total-NFE evaluations use 256. The source seed and initial-state pairing are fixed within each experiment and recorded in its protocol and result JSON. Spot uses one seed-42 policy and five paired 256-environment cells. The mechanism audit uses 128 environments, eight rollout steps, and 32 solver steps for each of four policies. The throughput audit uses ten warm-up calls followed by five rotated blocks of 20 synchronized calls per method.

A.4 COMMANDS

The paper figures and PDF can be rebuilt from the repository root with:

```
python3 paper/iclr2026/figures/gen_figures.py
latexmk -cd -pdf -interaction=nonstopmode \
  -halt-on-error paper/iclr2026/main.tex
```

The experiment-specific remote entry scripts are under `research_scripts/remote/`; each protocol records the seeds, checkpoints, environment counts, and pass/fail gates used before execution. A short validation run should precede any remote rerun, as documented by the repository workflow.

B ADDITIONAL NUMERICAL RESULTS

For the three new Go2 policies (seeds 43–45), the pooled stratified paired bootstrap gain of symmetric64 over IID-pair64 is 0.494, interval $[0.353, 0.642]$; the gain over random64 is 0.920, interval $[0.771, 1.075]$. These intervals summarize paired episodes within the three policies and are not substitutes for the policy-level uncertainty in the main text. Symmetric64 minus zero64 is -0.085 , interval $[-0.212, 0.041]$.

For the source-parity audit, the average single-source displacement RMS is 0.312, the average opposite-source displacement RMS is 0.311, the even RMS is 0.0386, and the odd RMS is 0.3088. Every policy passes the locked cosine, residual-ratio, exact-source-negation, finite-endpoint, and unchanged-parameter checks.

The Spot zero32-minus-zero64 difference is -0.110 , with paired bootstrap interval $[-3.730, 2.813]$. Thus reducing integration depth alone does not explain the gap between stochastic random inference and the deterministic controls.